The task, the rules, what was done, what was not, and the decisions that are the author's

Provenance document P0 of PROJECT spacetimes,
Revision_Deflation_CJT_and_Kohn-Sham_definition_C

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1. What this document is

This document is the account that the task specification requires of the work in the folder Revision_Deflation_CJT_and_Kohn-Sham_definition_C. It states

- the task, in the words of this provenance, with the author's hypotheses in his formulas, the instruction that created this folder, and how that instruction was read (Section 2);
- each of his hard rules and how it is met, with the place where that can be verified (Section 3);
- what the folder contains (Section 4) and what was done, in order (Section 5);
- the results in brief: what the definition by form C changes (Section 6);
- every **deviation** from the instructions, with its reason (Section 7);
- the **decisions that are the author's** to take or to confirm (Section 8), and the findings that concern his own documents and the earlier folders of the project (Section 9);
- what has **not** been done (Section 10);
- the inputs (Section 11) and the **log of errors** made during the work (Section 12).

The four provenance documents of the folder are called P0 (this one), P1 (the symbols), P2 (the calculations) and P3 (the two forms of the self-interaction), after the beginning of their file names. Every symbol that occurs here is explained in Section 13. Nothing in this document has to be looked up elsewhere; where it names another document of the folder, it does so to say what that document contains, not to define anything used here.

2. The task

2.1 The hypotheses

PROJECT spacetimes studies a sixteen-component Dirac field on two eight-dimensional spacetimes with the coordinates $x0, \dots, x7$: $x1, x2, x3$ are ordinary space, $x4$ is the ordinary time, $x0$ is a hidden space-like coordinate, and $x5, x6, x7$ are extra coordinates, time-like on the **4+4** spacetime and space-like on the **7+1** spacetime.

The author's hypotheses for this sub-project are:

1. **Deflation.** Each of $x5$, $x6$, $x7$ deflates exponentially. Flat spacetime is replaced by a pseudo-Riemannian space with a diagonal metric whose components along these coordinates are, with three positive rates $\mathrm{g55}, \mathrm{g66}, \mathrm{g77}$ and positive constants that may differ,

$$\text{4+4:}\quad g_{tt}^{(4+4)} = -\,\mathrm{const}_t^{\,2}\, e^{-2\,\mathrm{rate}_t\, x4}, \qquad \text{7+1:}\quad g_{tt}^{(7+1)} = +\,\mathrm{const}_t^{\,2}\, e^{-2\,\mathrm{rate}_t\, x4}, \qquad t = 5, 6, 7,$$

   where $\mathrm{rate}_5 = \mathrm{g55}$, $\mathrm{rate}_6 = \mathrm{g66}$, $\mathrm{rate}_7 = \mathrm{g77}$.

2. **The Dirac Lagrangians contain a mass term and a self-interaction.** The self-interaction is the quartic term of the author's document on the 7+1 problem,

$$\mathcal{L}_{\text{self}} = -\tfrac{1}{2}\sum_{r=0}^{8} g_r\, I_r, \qquad I_r = \sum_{|K|=r} (\psi^\dagger \text{Adj}\, \Theta_K\, \psi)(\psi^\dagger \text{Adj}\, \Theta^K\, \psi),$$

with nine couplings g_r, the 256 products Θ^K of Dirac matrices and the matrix Adj of the adjoint (Section 13).
3. **Two methods.** The effective potential for composite operators of Cornwall, Jackiw and Tomboulis (CJT) and the Kohn–Sham ansatz are both to be employed: to calculate additional predictions of the two models, and as a check on the existing computations. The author's solution of the 7+1 problem is to be carried over to the 4+4 problem.

2.2 The point at which the 7+1 construction does not carry over, and the author's decision

On the 7+1 spacetime the adjoint that makes the invariants I_r covariant is the adjoint that makes the energy positive: Adj $= \beta = \gamma^4$, $\bar\psi = \psi^\dagger\beta$. On the 4+4 spacetime the two are different matrices. The covariant adjoint is $\tilde\psi = \psi^\dagger C^-$ with $C^- = \gamma^0\gamma^1\gamma^2\gamma^3$, and the covariant Lagrangian $\tilde\psi(\gamma^\mu\nabla_\mu - m)\psi$ is indefinite: for fields that do not depend on $x5, x6, x7$, the only ones with a unitary evolution (the *quantized sector*), it is

$$\tilde\psi\,(\gamma^A\partial_A - m)\,\psi = -\psi^\dagger\, W_{1644}\,(i\partial_4 - \mathsf{h})\,\psi, \qquad W_{1644} = i\,C^-\gamma^4, \quad \mathsf{h} = \sum_{j=0}^{3}\alpha^j(-i\partial_j) + \beta\, m,$$

with $\beta = -i\gamma^4$, $\alpha^j = -\gamma^4\gamma^j$: the ordinary Dirac Lagrangian on the half of the field with $W_{1644} = -1$ and minus it on the half with $W_{1644} = +1$. The quantum theory of the free field is defined by the positive Lagrangian $\psi^\dagger(i\partial_4 - \mathsf{h})\psi$ for all sixteen components. With a self-interaction the quartic term can then be completed in two ways:

- **form C, the covariant form:** the invariants of the covariant Lagrangian, Adj $= C^-$, $I_r^{\text{C}} = \sum_{|K|=r} \text{sgn}_K\,(\tilde\psi\,\Theta^K\psi)^2$;
- **form Q, the quantized-sector form:** the invariants built with the adjoint of the positive Lagrangian, Adj $= \beta$, and with the matrices $\gamma^A_{44q} = -i\gamma^A$ ($A \le 4$), $\gamma^A_{44q} = \gamma^A$ ($A \ge 5$): $I_r^{\text{Q}} = \sum_{|K|=r} \text{sgn}^Q_K\,(\bar\psi\,\Theta^K_Q\psi)^2$.

They are related exactly, $I_r^{\text{C}} = \sum_{|K|=r} \text{sgn}^Q_K\,(\psi^\dagger W_{1644}\,\beta\,\Theta^K_Q\psi)^2$: form C is form Q with the matrix W_{1644} inserted in every bilinear.

The folder `Revision_Deflation_CJT_and_Kohn-Sham` of the repository adopted form Q, reported form C beside it for the couplings, for uniform matter and for pairs at rest, and put the choice before the author as his decision (its decision D1). **On 13 November 2026 the author decided: in a new folder form C is to be the definition; both options are to be recorded correctly and completely; and the options are to be explained and discussed in the documents and in the provenance files.** That instruction is the task of the present folder.

2.3 How the instruction was read

the instruction	what was done	where
a new folder	The folder `Revision_Deflation_CJT_and_Kohn-Sham_definition_C`, with its own program (`deflation_cjt_ks_c`), data, figures and documents. The folder `Revision_Deflation_CJT_and_Kohn-Sham` was not changed in any file; nor were the folder `Revision_Deflation` and the flat-spacetime part of the project.	the repository
form C is the definition	In every document of the folder the 4+4 model is defined by $\mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{mass}} + \mathcal{L}^{\text{C}}_{\text{self}}$: the positive kinetic and mass terms of the quantized sector with the quartic term of the covariant Lagrangian. Every result of the 4+4 model that depends on the self-interaction was computed again with form C: the mean-field couplings, uniform matter (now of two halves of the field), the effective potential and the vacuum, the pair couplings and the gap, the symmetries, the matter as a source of gravity, the thresholds with the Vorton-Scalar.	manuscripts; "dirac matrices deflating", Parts 5 and 6; `P3`
"C" with which kinetic term	The covariant kinetic term is indefinite and defines no quantum theory with states of positive norm (Section 2.2). Form C therefore means the covariant quartic term together with the positive kinetic term. The third possibility, the covariant kinetic term with the covariant quartic term, is recorded as a classical field equation, $i\partial_4\psi = \mathsf{h}\psi + W_{1644}\,\partial\mathcal{L}^{\text{C}}_{\text{self}}/\partial\psi^{\dagger}$; nothing is computed with it (Section 7, V8).	the same
both options recorded correctly and completely	Every calculation that involves the self-interaction on 4+4 is carried out for form C and for form Q with the same program, the same couplings of the Lagrangian and the same routines, and every table of such results prints both. The exact relation between the forms is verified for all 256 matrices in exact arithmetic. An independent script in another language repeats the comparison of the two forms, and deliberate errors introduced into it are detected.	`P3`; `P2`, Sections 5.1, 5.3, 5.4, 5.13, 5.14
explain and discuss the options in the documents and in the provenance files	A provenance document of its own, `P3`, written for a reader who knows neither form; a section "The two forms of the self-interaction: discussion" in the 4+4 manuscript; Section 6.7 of "dirac matrices deflating"; Sections 2, 6, 8 and 9 of this document; Section 3.2 of `P1` for the symbols; Section 2 of `P2` for the defining equations. Each discussion states what each form is, what speaks for it and against it, what does not distinguish the forms, and what could decide.	the documents named

2.4 The criticisms that the earlier task states, and what was done about each

The task specification of the sub-project (of which this folder is the third revision) states three criticisms of the first revision. They hold for this folder as they did for its predecessor.

criticism	what was done
The manuscripts are not useful for students: they send the reader to other manuscripts, documents, provenances or symbol files.	Each manuscript contains every symbol, definition and equation that it uses, with appendices for the symbols, the printed Dirac matrices, the curvature, and the programs and checks. A script searches every document for wordings that send the reader elsewhere (`check_self_contained.py`); none is found.
The manuscripts do not define the Dirac Lagrangians and Dirac equations correctly, because the self-interaction is missing.	The Lagrangians are $\mathcal{L}_{\text{kin}} + \mathcal{L}_{\text{mass}} + \mathcal{L}_{\text{self}}$ on both spacetimes, on 4+4 in both forms, and the field equations carry the cubic term; manuscripts, Secs. III and IV, and the document "dirac matrices deflating", Parts 3 to 6.
The same symbol is used for different objects.	A MASTER registry of 246 symbols with one meaning each, the list of the collisions found and their resolution, the symbols of the two forms, and eight machine checks: document `P1`. All documents of the folder use it.

3. The hard rules, and how each is met

rule	how it is met	where to verify
0. The audience is a group of clever physics Ph.D. students who do not know the solution; they must be able to learn from it and to teach it.	Every term and symbol is defined in words before it is used; derivations are given in steps that can be repeated by hand, and the text says how; each result names the check that verifies it.	any section of a manuscript, of “dirac matrices deflating” or of `P3`
1. All documentation, symbol and provenance files exist as .md, .tex and .pdf, complete, precise and correct.	Each of the eight documents of Section 4 exists in the three forms; the .tex and .pdf are built from the .md (for the manuscripts: the .md from the .tex) by scripts, and the build fails on an undefined reference or on a box wider than its space.	the files; `code/scripts/build_docs.sh`, `build_manuscripts.py`
2. Push to the repository as soon as an object is updated.	Every commit is carried at once to the author’s folder and fast-forwarded there; the push itself is done from the author’s computer (Section 7, V3). At the beginning of the work on this folder the rule was broken (Section 7, V9).	`git log`
3., 4. Working directory and repository.	The folder `Revision_Deflation_CJT_and_Kohn-Sham_definition_C` of the repository `https://github.com/once-ere/spacetimes`, in the working directory named by the rule.	
5. No symbol with two meanings; a MASTER provenance of the symbols, promulgated and used.	Document `P1` and the table `provenance/symbols_master.csv`; the checks MS1 to MS8 (8 of 8 pass) verify the registry, the documents and the columns of all data files against it. The first version of `P3` broke the rule (Section 12, N-E9).	`P1`, Section 5; `data/symbol_checks.txt`
6. Source code for every equation, result, figure and prediction; a figure for each prediction; commented code, instructions and provenance for every computed value; Rust preferred; the code executed.	One Rust program (36 source files, 17774 lines, 77 unit tests, 19 commands) computes everything; 38 figures are drawn from its data files; document `P2` gives, for every table and figure, the command, the source files and the data files, and prints the complete output of every check (310 checks of the program, all passing). Every command was executed in this folder; the reports in `data` are the outputs, and a check verifies that every data file was written here. Python is used for three purposes only (Section 7, V2).	`P2`
7. Never send the reader to another manuscript, document, provenance or symbol file; every equation, symbol and definition in the file itself.	Each document defines what it uses; `P3`, for example, contains the complete registry of symbols and the equations of gravity that its section on gravity needs. Program output printed in the documents uses the expression “the main project” for the flat-spacetime part of PROJECT spacetimes, whose program is used as a library and whose data files several checks compare against; the documents that print such output say so.	`data/self_contained_checks.txt`
8. Follow the instructions exactly, or explain in a provenance why not, and pause. No shortcuts.	Sections 7 to 10 and 12 of this document.	

4. What the folder contains

document (each as .md, .tex, .pdf)	content
`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Deflating Spacetime`	the manuscript for the 4+4 spacetime, in the style of Physical Review D; form C is the definition, every result of form Q is given beside it, and Sec. XVIII D discusses the two
`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 7+1 Deflating Spacetime`	the manuscript for the 7+1 spacetime, on which the self-interaction has one form
`dirac matrices deflating`	geometry, the sixteen Dirac matrices printed in full, spin connection, the Lagrangians with mass term and self-interaction, the field equations, the two forms of the 4+4 model and their algebra, their consequences in numbers and their discussion, solutions mode by mode; complete reports of the checks
`provenance/P0_task_rules_deviations_decisions`	this document
`provenance/P1_master_symbols_spacetimes`	the MASTER registry of symbols, the collisions and their resolution, the symbols of the two forms, the dictionary to the author's 7+1 document, the machine checks
`provenance/P2_calculations_code_and_checks`	every calculation: defining equations, commands, instructions, data files, figures, complete output of all checks, source files
`provenance/P3_two_forms_of_the_self_interaction`	the two forms from the beginning: the indefinite covariant Lagrangian, both definitions with their field equations, the exact relation, uniform matter, vacuum, pairs, symmetries, gravity, what has been established and what has not, the discussion
`README`	the list of contents and the instructions for building everything

other content	
`code/deflation-cjt-ks-c-rs`	the Rust program `deflation_cjt_ks_c`
`code/scripts`	independent verification in Python (two scripts and a test of them with deliberate errors); figures; the scripts that build and check the documents
`data`	126 data files written by the program, and the reports of all checks
`figures`	38 figures, each as .pdf and .png
`docs_src`	the sources of all documents

The two manuscripts are built from one source (`docs_src/manuscript`): the text that the two share is written once, and passages that differ are marked for one spacetime or the other, so that the manuscripts cannot drift apart.

5. What was done, in order

The folder began as a copy of the tracked files of the folder `Revision_Deflation_CJT_and_Kohn-Sham`. Nothing of that copy was committed before it had been produced again in this folder: every command of the program was executed here, every figure drawn here, every document rebuilt here.

commit	content
d392864	the program with form C as the definition: mean-field couplings of four densities for both forms, matter of the two halves, the CJT potential of two halves, the vacuum, both forms in the pair channels, matter of both forms as a source of gravity
6d654e8	correction of the search for the state of lowest energy per particle (Section 12, N-E2)
782315c, 85ca51a	data of the fourteen commands that do not involve the self-interaction, executed in this folder; identical in content to those of the folder with form Q
8277767	data of the commands `selfcoupling`, `kohnsham`, `pairing`, `gravity`
69d03e8	the ultraviolet criterion of the gap equation, and the upper limit of its momentum integral raised (Section 9, F6 and F7)
21f288b	data of the command `forms`: the two forms side by side (corrections N-E4 to N-E6)
9f8a206	`P3`, first version; the independent verification of the two forms; all figures
2ef3704	the symbols of the two forms entered in the MASTER registry; `P3` rewritten with them; the names of the program and of the data columns changed accordingly (N-E9, N-E10)
0cd79e2	“dirac matrices deflating” with form C as the definition
2b81eb2	the test of the independent scripts with deliberate errors
25d5665, a824eab	the two manuscripts; corrections after reading the typeset pages (N-E15, N-E16)
the commits that follow	`P1`, `P2`, `README`, this document; the check that every data file was written in this folder

6. The results in brief: what the definition by form C changes

Each statement is verified by the checks named in the manuscripts, in `P3` and in `P2`. Numbers are in units of the fermion mass m unless said otherwise. The sets of couplings are **set A**, a scalar attraction alone, $g_0 = -g$, and **set B**, $g_0 = -g$ with a vector repulsion one tenth as large, $g_1 = +0.1\,g$, with a strength $g > 0$ that grows as e^{τ_d} along the deflation clock τ_d.

What is the same in both forms. Everything that does not involve the contact self-interaction: the geometry, the Dirac operator, the solutions mode by mode, the exchange of the Vorton-Scalar (whose Yukawa coupling contains $\bar{\psi}$ in both forms), the cosmological part. The reduction of the nine couplings to four effective couplings g_0^{eff}, g_1^{eff}, g_7^{eff}, g_8^{eff}, with the same Fierz matrix. The interaction between two particles of the same half of the field. The equivalence of the Kohn–Sham method and the CJT effective potential, which agree for the two halves in both forms to 4.9×10^{-11} on 80 states. On the 7+1 spacetime there is one form.

The mean field. The energy density of the self-interaction in uniform matter is, in both forms, a sum of four squares, of the total number density and scalar density and of the differences of these densities between the two halves; the four coefficients are the same four combinations of the effective couplings, and the two forms exchange the coefficients of the totals with those of the differences. For set A the attraction $\frac{15}{16}g_0$ acts on the total scalar density in form Q and on the difference of the scalar densities in form C. Between two particles of the same half the coupling is $\frac{7}{8}g_0$ in both; between particles of different halves it is g_0 in form Q and $-g_0$ in form C.

Matter. In form C matter polarizes: the state of lowest energy is polarized in 48 of the 48 cases examined in form C and in 0 of them in form Q. Self-bound matter of form C has one half of the field filled, that of form Q both halves equally. The strength $g\,m^{D-1}$ at which a gas becomes self-bound matter is, in $D = 4$ space dimensions with momentum, 224.74 (form C) against 82.11 (form Q)

for set A and 332.29 against 121.09 for set B; in $D = 3$, 24.572 against 9.809 and 31.167 against 12.382: form C needs 2.51 to 2.74 times the strength and binds later on the clock by 0.92 to 1.01. A gas of fixed density polarizes before it binds.

Vacuum. With the filled states of negative energy below a cutoff Λ a field without mass term acquires a mass above the same critical strength in both forms (31.583 in units of Λ^{-3} for $D = 4$, set A), and of the same magnitude; in form Q the two halves get the same mass, in form C masses of opposite sign, which is the mass term $\tilde{\psi}\psi$ of the covariant Lagrangian. With a bare mass form Q passes smoothly through the critical strength, and form C has a sharp transition at which the symmetry between the halves breaks.

Pairs. A fermion pair along one of the space-like axes $x0, \ldots, x3$ consists of one particle of each half, a pair along one of the additional times $x5, x6, x7$ of two particles of the same half. In form C the contact couplings therefore enter the gap equation of the pair along the hidden axis with the sign opposite to that of form Q (amplitudes compared at random momenta, 2.2×10^{-15}): a scalar contact attraction lowers its gap, a vector repulsion raises it, and the four pairs along $x0, x5, x6, x7$, which are degenerate in form Q, are not. At the clock reading 2.04 of the example ($m_{_s} = m$, Fermi momentum $0.5\,m$) the exchange of the Vorton-Scalar alone gives the gap 0.0247 of the Fermi kinetic energy; the scalar contact attraction of the example multiplies it by 12.1 in form Q and removes it in the pair along the hidden axis of form C.

Symmetries. The self-interaction is chirally invariant if $g_1^{\text{eff}} - g_7^{\text{eff}} = 0$ in form C and if $g_0^{\text{eff}} + g_8^{\text{eff}} = 0$ in form Q. Every self-interaction of form C that does not vanish breaks the internal symmetry $SO(4) \times U(1)_E$ of fields in ordinary space, on which the hypothesis about the Higgs field rests, down to the rotations of $x5, x6, x7$; form Q keeps $SO(4)$ for all couplings.

Gravity. With self-bound matter of set B at three times its binding threshold as the source of the tuned Einstein–Lovelock equations (the gravity in which the undisturbed deflation is a solution without matter), and with the smaller of the two amounts of matter examined, the rate of deflation at the end of the evolution is 34.1 times the initial rate for the matter of form C and 15.6 times for that of form Q; the conclusions (matter accelerates the deflation; the undisturbed deflation is unstable) are the same.

A result that concerns both forms and both spacetimes: the ultraviolet criterion. The gap equation has a solution that does not depend on the upper limit of its momentum integral only if the ultraviolet strength of its kernel, U_{uv}, is below $4\pi^2$ (Section 9, F6 and F7).

7. Deviations from the instructions, and the reasons

V1. The five supplements of the first revision are not repeated as separate documents. Hard Rule 7 forbids a document to send its reader to another, and a supplement is a document to which a manuscript sends its reader. The content of the supplements is therefore inside each manuscript:

supplement of the first revision	where its content is in each manuscript of this revision
Vorton-Scalar and superfluid	Secs. VI, VIII, X, XI, XIV
effective action along the clock	Sec. XII
cosmology	Sec. XVI A to D, F, I to K
microwave background	Sec. XVI E, G, H
Higgs hypothesis	Sec. XVII

The derivations are in the sections named. Of the tables of the supplements, the sections print those that carry the argument and quote numbers from the others; the others are printed in full in the appendix “Complete tables of the calculations” of each manuscript, each with a caption that defines its columns.

V2. Python is used beside Rust, for three purposes: (i) the independent verification, which must not share code with the program that it verifies and therefore uses another language and other libraries (computer algebra, a general-purpose integrator, other special functions); (ii) the figures; (iii) the scripts that assemble, typeset and check the documents. No result of a manuscript comes from Python alone. Mathematica was not available.

V3. The push to GitHub is done from the author's computer. The work is carried out in a workspace without the author's credentials: a push from the workspace is refused by the server, and none is possible from the part of his computer that the work can reach. Each commit is transferred at once to his folder as a git bundle and fast-forwarded there; the folder is then ahead of the remote until he runs `git push origin main` in it.

V4. The test with deliberate errors does not cover the whole of the first independent script. It introduces 45 errors (29 in the script for the self-interaction, the Kohn-Sham and CJT calculations, pairing and gravity, 16 in the cosmological part of the other script); all are detected. The part of the first script that verifies the geometry, the Dirac operators and the modes has no such test.

V5. The titles of the manuscripts are those of the first revision; the task names no titles.

V6. An external Einstein–Boltzmann code was not used as a cross-check. The spectra of the microwave background are computed with the program of this project, whose version for static extra coordinates had been validated in the flat-spacetime part of the project; no external code was built or run in this sub-project.

V7. The 7+1 manuscript is built in this folder too, although the instruction concerns the 4+4 model. One source builds both manuscripts, and the finding on the upper limit of the gap equation (Section 9, F6) concerns both: the 7+1 manuscript of this folder contains the ultraviolet criterion and gaps computed with the higher limit, which differ in the last digits from those of the folder with form Q. In everything else it has the content of its predecessor.

V8. Nothing is computed with the third option, the covariant quartic term with the covariant (indefinite) kinetic term. Its field equation is derived and recorded in every document that defines the forms. It conserves $\psi^\dagger W_{1644}\psi$ and not $\psi^\dagger\psi$, and it is not the equation of a Hamiltonian with states of positive norm when both halves of the field are occupied; no quantum theory of particles, and therefore no matter, vacuum or pairing, can be computed from it with the methods of this project.

V9. Hard Rule 2 was broken at the beginning of the work on this folder. From the instruction (14:06, Pacific time) to the first commit of the folder (15:34) nothing was committed or carried to the author's folder, while the program was being extended and the long commands were running. The author pointed it out. Since then every step ends with a commit that is transferred at once (Section 12, N-E1).

V10. The top-level `README.tex` and `README.pdf` of the repository were not rebuilt after the author changed a word in the top-level `README.md` (his commit `43e4525`, "Licence" to "License"): the flat-spacetime part of the project, to which these files belong, is paused by his instruction. His change was merged; the three files differ in that one word until that part is taken up again.

8. Decisions that are the author's

D1. The form of the self-interaction on the 4+4 spacetime: decided. Form C, by the author's decision of 13 November 2026 (Section 2.2). Two matters connected with it remain his:

- *Which document stands for the 4+4 model.* The repository now holds two 4+4 manuscripts with the same title, one in the folder with form Q as the definition and one in this folder. If this folder supersedes the other, the other should say so; it was not changed.
- *The sets of couplings.* The numbers of this folder are given for the sets A and B. Set B has the ratio of the vector to the scalar coupling of the author's reference example. The formulas hold for all couplings; other sets are a matter of running the command `forms` with them.

D2. The bare mass. With the Vorton-Scalar, a bare mass m_D breaks the chiral symmetry explicitly and gives the phase mode a mass. Results that rest on a massless phase mode (the stepped fluid of the cosmological part) presuppose $m_D = 0$ or a mass of the phase mode below the temperatures concerned. Which is meant is the author's decision.

D3. The couplings of the reference example. See Section 9, F2: are -10^5 and $+10^4$ the couplings g_0, g_1 of the Lagrangian or the couplings G_S, G_V of the mean-field energy?

D4. The symbols of the metric. The metric of this sub-project is written $g^{(4+4)}$, $g^{(7+1)}$ in the author's sign convention (space-like components positive), and the rates keep his names g55, g66, g77, upright. The names of the first revision for a metric of the other sign convention are retired.

D5. Open from the first revision: the assumption about the perturbations of ordinary space and its closures; the nature of whatever sustains the deflation; the choice between Einstein gravity with such an agent and Einstein–Gauss–Bonnet gravity.

D6. The earlier folders and the upper limit of the gap equation. The folders `Revision_Deflation_CJT_and_Kohn-Sham` and `Revision_Deflation` and the flat-spacetime part of the project solve the gap equation with an upper limit of the momentum integral that is too low (Section 9, F6). They were not changed: the first by the instruction that created this folder, the others because they are finished or paused. Whether they are to be computed again with the higher limit is the author's decision.

D7. The hypothesis about the Higgs field under form C. The pattern on which the hypothesis rests, a doublet under a hidden $SO(4)$ that the condensate along the hidden axis breaks to $SO(3)$, is present in the model as now defined only if there is no contact self-interaction at all (Section 9, F5). Whether the hypothesis is to be kept for the model without contact self-interaction, tied to form Q, or given up, is the author's decision.

9. Findings that concern the author's own documents and the earlier folders

F1. Only four of the nine couplings are independent. The identity $(1 + \mathsf{F})I = 0$ holds for anti-commuting fields with the Fierz matrix of his document, in both forms. A tensor coupling $g_2, \dots, g_6$ is not a new parameter.

F2. The couplings of the Kohn–Sham example are not the Hartree–Fock couplings of $g_0 = -10^5$, $g_1 = +10^4$. His document states the Hartree–Fock couplings of these values correctly as -98750 and $+20000$ (check `self-X1`), and its Kohn–Sham calculation uses $G_S = -10^5$, $G_V = +10^4$. With the first pair, matter on flat 7+1 spacetime is not self-bound: its energy per particle has only a local minimum above the mass. The numbers of his document are reproduced with the second pair. If the mean-field couplings are meant to be the Hartree–Fock couplings, the Lagrangian couplings that give his example are $g_0 = -105341.6$ and $g_1 = +2484.5$ in the same units.

F3. The coupling of the example exceeds the critical coupling of the vacuum when the filled states of negative energy are included with a cutoff above 0.6665: the Kohn–Sham calculation without those states presupposes a cutoff below that value. The same holds for self-bound matter of the 4+4 model by contact couplings alone, in both forms and more so in form C, which needs the stronger couplings: at each binding threshold the largest cutoff for which the vacuum is stable is below the mass of the particles.

F4. The deflating spacetime is an unstable solution of the gravity in which it is a vacuum solution, and in Einstein gravity it needs a source without energy density and with a tension; Kohn–Sham matter cannot be that source, in either form.

F5. Form C breaks the internal symmetry on which the hypothesis about the Higgs field rests. In form Q, and on 7+1, a vector or axial-vector coupling breaks $U(1)_E$, the second phase symmetry of the fields in ordinary space, and every coupling keeps the rotations $SO(4)$ of the four extra axes. In form C every self-interaction that does not vanish breaks the three rotations that

mix the hidden axis with $x5, x6, x7$ and breaks $U(1)_E$ (check `44c-S12`, exact): the transformations generated by $\gamma^0\gamma^t$ are boosts of the 4+4 frame, and the unitary transformations generated by the same matrices are not elements of the frame group, so that a covariant quartic term has no reason to be invariant under them. With an attractive scalar contact coupling the pairs along the additional times, not the pair along the hidden axis, have the larger gap.

F6. The gap equation of the earlier calculations was solved with an upper limit of the momentum integral that is too low. The folders named in D6 integrate to 60 times the larger of m and k_F. The kernel falls only as the inverse square of the momentum, so that the gap function has a tail that falls as a power. Measured in this folder with the limits 60, 2×10^3, 2×10^4 and 2×10^5 on 192 kernels (check `pc-P7`): with the limit 60 the gaps of the exchange of the Vorton-Scalar alone are too small by up to 1.08 per cent at the strongest coupling of the histories, and gaps with attractive contact couplings by more, the more the nearer the kernel is to the criterion of F7 (the complete table by classes of the ultraviolet strength is in the manuscripts and in `P3`). Quantities derived from the gap function change with it: in the comparison of the Ginzburg–Landau coefficients with their weak-coupling formulas the largest deviation at the third moment is 9.8 per cent with the higher limit, where the limit 60 had given 7.9 per cent (check `def-P3`). This folder uses 2×10^4 throughout, at which the gaps below the criterion have converged. The conclusions of the earlier folders are not affected; their last digits are.

F7. The gap equation of the model has no solution beyond a definite coupling. A solution that does not depend on the upper limit exists only if the ultraviolet strength of the kernel,

$$U_{\mathrm{uv}} = 2\,u_S\,g_{VS}^2 - s_{\mathrm{ct}}\,m_{\mathrm{ct}}^2\big[(g_0^{\mathrm{eff}} - g_8^{\mathrm{eff}})\,u_S + (g_1^{\mathrm{eff}} + g_7^{\mathrm{eff}})\,u_V\big], \qquad u_S = \frac{\pi^2}{8} - \frac{1}{2}, \quad u_V = \frac{3\pi^2}{4} - 1,$$

is below $4\pi^2$; above it the gap grows in proportion to the upper limit (the pair collapses to short distances, as two relativistic particles do in a Coulomb potential that is too strong). The exchange of the two modes of the Vorton-Scalar alone reaches the criterion at $g_{VS}^2 = 26.90$, the clock reading 3.29, beyond the last reading 2.77 of the histories of the project; attractive contact couplings bring it nearer. No entry of the gap tables of the earlier folders lies above the criterion; the entries nearest to it, those with the scalar contact attraction at the last clock reading, are the ones that the limit 60 changes most (F6). In this folder the contact couplings enter the pair along the hidden axis of form C with the opposite sign, a vector repulsion then acts as an attraction, and with the vector coupling of the examples the criterion is reached at the clock reading 2.46; no gap is given above the criterion. Along the deflation clock every coupling grows without bound, so that the criterion is reached at a definite moment in every history: beyond it the model needs physics at short distances that it does not contain.

10. What has not been done

1. *Which form is the right one.* No calculation of this folder decides between form C and form Q. The documents set out what speaks for each and what could decide: a principle (a covariant theory on the full 4+4 spacetime, or a theory of eight equivalent internal states), a consistent quantization of the indefinite covariant Lagrangian, or an observable consequence.
2. *States with coherence between the two halves of the field.* The density matrices of this folder are diagonal in W_{1644}. Because the interaction connects the halves, states with a coherence between them are possible in principle, in both forms; whether one has a lower energy has not been examined.
3. *Other couplings.* The numbers are for the sets A and B; the exact formulas hold for all couplings.
4. *The continuation of form Q with a mass that has changed its sign.* Where the gap equations of form Q have no solution with non-negative masses (108 of 3936 states examined, all with unequal halves), the continuation was not followed.

5. *The potential of the order parameter with the contact couplings of form C*, and *pairing inside polarized matter*: in matter of form C with one filled half a pair along the hidden axis, which needs one particle of each half, cannot form at all; the gaps are computed for particles of the mass m without regard to the polarization.
6. *Matter beyond the mean field; pairing inside dense self-bound matter; a self-consistent solution with the created particles acting on the metric.*
7. *A test with deliberate errors for the geometric part of the first independent script* (Section 7, V4); *separate supplements* (V1).
8. *The earlier folders were not corrected* for the upper limit of the gap equation (Section 8, D6), and *the paused stages of the flat-spacetime part of the project* were not touched (Section 7, V10).

11. Inputs

input	what was taken from it
the author's task specification for the sub-project (private; not in the repository)	the hypotheses and the rules of Sections 2.1 and 3
the author's instruction of 13 November 2026	the task of this folder (Section 2.2)
the folder `Revision_Deflation_CJT_and_Kohn-Sham` of the repository	the program, the scripts and the sources of the documents from which this folder started; it was not changed
his document `KOHN_SHAM_GRAVITY` (folder `7+1` on his computer; its .tex source was read in full)	the definition of $\mathcal{L}_{\text{self}}$, of Θ^{K}, I_r, g_r; the Fierz matrix; the Kohn–Sham equations and the reference numbers; the Einstein–Lovelock equations for two scale factors and the reference evolution
his document `DIRAC16COMPLEX` (same folder)	the critical coupling of the vacuum in seven space dimensions
the notebooks and papers of the project, as recorded in the flat-spacetime part	the two sets of real 16×16 Dirac matrices (files `data/gamma1644.txt`, `data/gamma1671.txt` at the top of the repository); the conventions of the fields
the folder `Revision_Deflation` (the first revision)	the calculations that are repeated here; it was not changed

The author's files were read, not changed, and none of them is copied into the repository.

12. Log of errors

Every error made during the work is recorded here, including errors that were corrected before anything was committed. Section 12.1 lists the errors made in the work on this folder. Section 12.2 repeats the log of the folder `Revision_Deflation_CJT_and_Kohn-Sham`, from which this folder started and whose program and documents it inherited with their corrections.

12.1 Errors made in the work on this folder

No.	what was wrong	how it was found	correction
N-E1	**Hard Rule 2 was broken.** From the author's instruction to the first commit of the folder, about an hour and a half, nothing was committed or carried to his folder.	The author pointed it out.	Every step ends with a commit that is transferred to his folder at once (Section 7, V9).

No.	what was wrong	how it was found	correction
N-E2	The first version of the search for the state of lowest energy per particle refined only around the lowest value of a coarse scan of the density. A narrow bound minimum between two scan points was missed, and binding thresholds came out up to 2.7 per cent too high.	The command `forms` prints beside every threshold the threshold of the routine for matter of one kind; the two disagreed.	Commit `6d654e8`: a finer scan, and a refinement around every local minimum.
N-E3	The folder began as a copy of another folder, so that its folder `data` at first held the data files of the other program; a file that no command of this folder wrote again would have carried the numbers of the other program into the documents.	Foreseen when the folder was created.	No file of the copy was committed before a command of this folder had written it again. The script `check_all.sh` now verifies that every file in `data` names this folder in its head.
N-E4	Check `fm-F3` required a state at every polarization. In form Q, at three and six times the threshold and at high density, 108 of 3936 states with unequal halves do not exist with non-negative masses.	The first complete run of the command `forms` reported six of nine checks.	The check states the fact and requires the states with equal halves and with one filled half.
N-E5	Check `fm-F4` covered clock readings up to 6. In form Q the state with equal halves reaches the eigenvalue 1 of the matrix $\mathsf{R}_{\mathrm{gap}}$ at the clock reading 5.30 ($D = 3$, set A), and states with slightly unequal masses appeared in the table.	The same run.	The table ends at the clock reading 5 and carries the eigenvalue as a column; the check reports where each case reaches 1.
N-E6	Check `fm-F5` required that the pressure of a gas of fixed density become negative in every case. In form C at the higher density it stays positive in 3 of the 8 cases.	The same run.	The check states what happens in each case.
N-E7	The vacuum of the two halves has two solutions that differ by the exchange of the halves. The first version of the command printed whichever the solver found, and the first version of the independent check compared the signed masses.	The independent check failed on a case in which the two programs had found different members of the pair.	The program prints a fixed representative (the half with the positive mass first); the independent check compares the magnitudes and the relative sign.
N-E8	The gap equation was solved with the upper limit 60 of its momentum integral, taken over from the earlier programs of the project; the gaps depend on it.	The gap solver of the independent script used another limit and disagreed with the program.	Commit `69d03e8`: the ultraviolet criterion, the limit 2×10^{4}, check `pc-P7` (Section 9, F6 and F7).

No.	what was wrong	how it was found	correction
N-E9	**Hard Rule 5 was broken.** The first version of `P3` (commit `9f8a206`) introduced symbols that had not been compared with the MASTER registry; fifteen groups of them collided, or nearly collided, with registered symbols.	Comparison of `P3` with the registry before the next document was written.	Commit `2ef3704`: the symbols entered in the registry, `P3` rewritten, check MS8; `P1`, Section 3.3, lists what was replaced.
N-E10	A column of the data files had a name that is a symbol of one of the author’s notebooks (check MS3 failed).	The check.	The eight names of the mean-field couplings in the program and in the data files were changed; the four commands concerned were executed again; every number is unchanged.
N-E11	While `P3` was being rewritten: a script changed the registry in memory and did not write the file; a replacement of the name of a matrix throughout the text also changed a sentence about the letter itself; a test for leftover symbols gave a false alarm on an index list.	Reading the saved files and the differences.	Corrected before the commit.
N-E12	An equation of `P3` was wider than the page. A line of a report of checks was too long for the typesetting program, which stopped. The script `check_all.sh` then compared the printed reports with the unfolded files and called two documents outdated that were not.	The builds; the script.	The equation is set on two lines; the script that assembles the documents continues a report line of more than 1800 characters on the next line; `check_all.sh` folds in the same way before it compares.
N-E13	Drafts contained a number that no program states (in a reworded sentence), a section number that the number check took for a result, the name of a generated text that does not exist, and a reference to an appendix by a number that had changed.	The number check; the script that assembles the documents.	Corrected before the commits.
N-E14	One transfer of a commit to the author’s folder failed half-way: the permission to replace files there had lapsed, and the program git left a lock file and two files of the new commit as untracked files.	The output of the transfer.	The permission was renewed by the author; the lock was removed; the two files were compared by their hashes with those of the commit, found identical, and removed; the transfer was repeated. No file of the author’s was touched.
N-E15	A generated text counted the evolutions with gravity whose rates run away over all sixteen evolutions of the folder, where the sentences that use it speak of the eight of one spacetime.	Writing the section on gravity of the manuscripts.	A count for each spacetime.

No.	what was wrong	how it was found	correction
N-E16	**A document with two outdated numbers was committed.** Commit `25d5665` of the manuscripts quoted, in the section on the effective action, two numbers of the comparison of the Ginzburg–Landau coefficients that belong to the upper limit 60 (1.0 and 7.9 per cent; the program of this folder states 1.2 and 9.8). The sentence had been typed by hand in the predecessor of this folder.	The number check no longer found the second number among the numbers that the programs state.	Commit `a824eab`. Then every number that the programs of the folder with the limit 60 state and those of this folder do not state was searched for in the sources of all documents of this folder: no other occurrence.
N-E17	In the same commit the table of the checks in both manuscripts gave 7 of 8 for the checks of the symbols, because `P1`, `P2` and a status file of the folder still contained replaced symbols. The message of the commit says so.	Known when it was committed.	The documents were rebuilt when `P1` and `P2` were complete, and the status file was removed.
N-E18	The 7+1 manuscript said that gaps computed with the limit 60 are too small “by the percentages” of a table whose classes also contain kernels of the sign that does not occur on the 7+1 spacetime.	Reading the typeset page.	“By up to”; the caption says that the classes hold the kernels of both signs.
N-E19	The first build of the manuscripts had seven boxes wider than their space and one table of symbols taller than a page.	The build script, which fails on them.	Equations and tables rearranged; the table of symbols divided in two.

The cause of N-E16, and the remedy. The number check accepts a number of the text if any program states that number anywhere; it cannot know which statement a sentence means. A number typed by hand can therefore survive a change of the calculation behind it. The remedy used here is the comparison described in the row N-E16. Where a sentence quotes a result of a calculation that may change, the manuscripts of this folder take the number from a generated text and not from the keyboard; the sentence of N-E16 was one of those that had been typed.

12.2 Errors of the work that this folder inherited

These errors were made, and corrected, in the folder `Revision_Deflation_CJT_and_Kohn-Sham`; the commits named are commits of that work. For the errors C-E1 to C-E11, which were corrected before anything was committed, the record kept during the work names the error and what exposed it; an entry says no more than the record contains.

No.	what was wrong	how it was found	correction
C-E1	The first formula for the pair couplings acted on pair states without projecting on the states of positive energy and gave other integers.	Comparison with the exact diagonalization for two particles at rest.	The projected map; checks `71s-S9`, `44q-S9`.

No.	what was wrong	how it was found	correction
C-E2	An early text called the author's values G_S, G_V of the Kohn–Sham example the Hartree–Fock couplings of g_0, g_1 with the same numbers.	The exact Hartree–Fock formula (check `self-X1`).	The texts distinguish the two; finding F2.
C-E3	In the first version of the check of the mass of the phase mode (`uni-U3`) a deviation appeared that was an artefact of the momentum cutoff.	The check failed.	Both sides of the comparison are evaluated with the same cutoff.
C-E4	In the history of deflating matter (`ks-K5`) a minimization ran without bounds.	Unphysical values in the output.	The search is bounded.
C-E5	An angular average of the pairing amplitudes evaluated a function outside its domain.	The program stopped.	Corrected.
C-E6	The tolerance of an independent comparison of two integrations of the deflating spacetime with matter (`Icks-G5`) was set tighter than the accuracy of the comparison.	The check failed although both integrations were correct.	The tolerance follows from the measured accuracy, which the check states.
C-E7	Labels of several new figures collided with curves, and some used the names of the program or of the author's document instead of the registered symbols.	Looking at each figure.	Figures redrawn.
C-E8	The registry of symbols at first contained collisions introduced by this sub-project: one letter for a velocity dispersion and for a self-energy, one for a pair state and for a pair correlation, and two definitions of x.	The checks of the registry and reading the manuscripts.	New symbols σ_v, $\Phi(\vec{p})$; one definition of x.
C-E9	A generated table contained section numbers typed by hand.	Reading the built manuscript.	The table carries references that the typesetting resolves.
C-E10	The description of the test with deliberate errors repeated a statement that belonged to a test of the first revision.	Comparison with the report of the script of this folder.	The text states what the report of this folder says.
C-E11	A clause about the degeneracy of pair states in the section on the Higgs hypothesis had not been verified.	Review of the section against the checks.	Removed.

No.	what was wrong	how it was found	correction
C-E12	**A failing check was committed.** The message of commit `a232561` says that the symbol checks pass `7 of 7`; the report in that commit says `6 of 7`. With the documents `P1` and `P2` in the folder, check MS4 failed: two generated texts wrote the rates as plain text.	Reading the whole report instead of its last lines, one step later.	Commit `990d697`: both texts write the rates upright.
C-E13	**A second failing check was committed.** Since `P2` was added, the check of the numbers failed (it did not recognize `Section 5.11` as a section number, and two run times had no entry in its lists), and `P2` printed the failing report.	As C-E12.	Commit `8231b87`.
C-E14	`P2` printed a caption with an unresolved placeholder.	The new script `check_all.sh`.	Commit `8231b87`.
C-E15	The running header of `P1` and `P2` named the folder of the first revision.	Looking at the typeset pages.	Commit `56f9162`.
C-E16	In the figure of the creation of particles the labels of two axes touched, and two legends lay on curves; the figure is in the 7+1 manuscript.	Looking at the typeset page of the new document.	Commit `56f9162`: labels on two lines, legends moved, manuscript rebuilt.
C-E18	The headers of two generated tables carried equation numbers of documents of the first revision.	Writing the appendix of complete tables.	The headers describe the columns; the manuscripts refer to their own equations.
C-E19	In eleven of the thirty figures a label or a legend touched a curve, a marked line or the frame (figures of the bound pair, of the effective action, of the vacuum mass, of the extinction, of the geodesics, of the two evolutions with gravity, of the Hubble constant, of the quench, of the scalar modes, of the forecast, and of superfluid dark matter); and three figures wrote the coordinates $x0, \dots, x3$ with two dots. Most of these figures had been taken over from the first revision without being looked at again.	Every figure was looked at, one by one, after C-E16.	Labels and legends moved, lines shortened, captions adjusted where a label was shortened; all documents rebuilt.
C-E17	A sentence drafted for “dirac matrices deflating” took the roots of the constraint of the gravitational equations for rates of growth.	Reading the statement of check `gr-G5` before the document was committed.	The document says what the check says.

The cause of C-E12 to C-E14, and the remedy. After long builds only the last lines of the

outputs were read, and a verdict further up was missed. The script `code/scripts/check_all.sh` now runs the checks of the documents, reads the verdict of every report in `data`, verifies that no built document contains an unresolved placeholder and that every report printed in a document is the current one, and prints `ALL PASS` only then. Nothing is committed without it.

13. The symbols used in this document

symbol	meaning
$x0, \dots, x7$; xt	the eight coordinates ($x4$: ordinary time); a deflating coordinate, $t = 5, 6, 7$
g55, g66, g77; rate$_t$; gsum	the three rates of deflation (the author's names; upright, never with a subscript); the rate of xt; their sum
const$_t$	the positive constants of the hypothesis, one for each deflating coordinate
$g^{(4+4)}_{\mu\nu}$, $g^{(7+1)}_{\mu\nu}$	the metrics of the two deflating spacetimes, space-like components positive
V_d	$\sqrt{\lvert\det g\rvert}$, the proper volume of a unit coordinate cell of $x5, x6, x7$; proportional to $e^{-\mathrm{gsum}\, x4}$
τ_d	the deflation clock, gsum $(x4 - x4_{\mathrm{ref}})$, with a reference time $x4_{\mathrm{ref}}$
ψ; $\hat{\psi}$	the sixteen-component Dirac field; the weighted field $\sqrt{V_d}\,\psi$
γ^A; γ^μ; ∇_μ	the eight real 16×16 Dirac matrices of an orthonormal frame, $A = 0, \dots, 7$; the same with a coordinate index; the covariant derivative of a spinor
β; $\bar{\psi}$; α^j; h	γ^4 on 7+1 and $-i\gamma^4$ on 4+4; the adjoint $\psi^\dagger\beta$; on 4+4 the matrices $-\gamma^4\gamma^j$, $j = 0, 1, 2, 3$; the Hamiltonian matrix of a free particle
C^-; $\tilde{\psi}$; W_{1644}	on 4+4: the matrix $\gamma^0\gamma^1\gamma^2\gamma^3$; the covariant adjoint $\psi^\dagger C^-$; the matrix $iC^-\gamma^4$, Hermitian with the eigenvalues $+1$ and -1, eight times each; its two eigenspaces are the two *halves* of the field
Adj	the matrix of the adjoint in a bilinear $\psi^\dagger \mathrm{Adj}\,\Theta^K\psi$: β on 7+1 and in form Q, C^- in form C
form C, form Q	the two forms of the self-interaction of the 4+4 model in the quantized sector: with the covariant adjoint (the definition) and with the adjoint of the positive Lagrangian (the alternative)
γ^A_{44q}; Θ^K_Q; sgn$_K$, sgn^Q_K	the Dirac matrices of form Q, $-i\gamma^A$ for $A \le 4$ and γ^A for $A \ge 5$; their products with a phase that makes the bilinears real; the signs ± 1 of the squares of Θ^K and of Θ^K_Q
I^{C}_r, I^{Q}_r; $\mathcal{L}^{\mathrm{C}}_{\mathrm{self}}$, $\mathcal{L}^{\mathrm{Q}}_{\mathrm{self}}$	the nine invariants and the self-interaction in the two forms
g^{eff}_0, g^{eff}_1, g^{eff}_7, g^{eff}_8; g; set A, set B	the four independent combinations of the nine couplings (scalar, vector, axial vector, pseudoscalar); the strength of the couplings; the two sets of couplings $g_0 = -g$, and $g_0 = -g$ with $g_1 = +0.1\,g$
R$_{\mathrm{gap}}$	for matter of the two halves, the matrix of the derivatives of the right-hand sides of the two gap equations with respect to the two masses
s_{ct}; m_{ct}; g_{VS}	the sign ± 1 with which the contact couplings enter the pairing potential of a pair; their range parameter; the coupling of the Vorton-Scalar to the fermions
U_{uv}; u_S, u_V	the ultraviolet strength of the kernel of the gap equation; its two numerical constants

symbol	meaning
D; τ_d in the tables of Section 6	the number of space dimensions in which the particles have momentum (4, or 3 for fields in ordinary space); the deflation clock
m; m_D	the mass of the fermion; the part of it that is a bare mass in the model with the Vorton-Scalar
$\mathcal{L}_{\text{kin}}$, $\mathcal{L}_{\text{mass}}$, $\mathcal{L}_{\text{self}}$	the three parts of the Lagrangian: kinetic term, mass term, self-interaction
K; r; Θ^K, Θ_K	a strictly increasing list of indices out of $0, \dots, 7$; its length, the rank; the product of the Dirac matrices of the list, with a phase that makes the bilinears real, and its inverse
I_r; g_r	the invariant of rank r of the covariant Lagrangian; the nine couplings, $r = 0, \dots, 8$ (g_0 scalar, g_1 vector)
F	the Fierz matrix, a 9×9 matrix of rational numbers with $\mathsf{F}^2 = 1$
G_S, G_V; G_S^{HF}, G_V^{HF}	the scalar and the vector coupling of the mean-field energy of uniform matter; their values from the nine couplings in the Hartree–Fock approximation
k_F	the Fermi momentum
Λ	a momentum cutoff
$SO(4)$; $U(1)_E$	for fields that depend on $x1, x2, x3, x4$ only: the rotations of the four extra axes $x0, x5, x6, x7$, generated by the products $\gamma^a\gamma^b$ of their Dirac matrices; a second phase symmetry, generated by a multiple of $\gamma^0\gamma^5\gamma^6\gamma^7$
Vorton-Scalar	the complex scalar field of the model, coupled to $\bar{\psi}\psi$; its radial mode and its phase mode are its two kinds of excitation